

SRINIVAS UNIVERSITY
INSTITUTE OF ENGINEERING & TECHNOLOGY



(SUBJECT: ARTIFICIAL NEURAL NETWORKS)
(SUBJECT CODE:24SBT113)

AN INDIVIDUAL TASK REPORT ON
“EQUATIONS OF ALL MODELS”

Submitted in the partial fulfillment of the requirements for the Fourth semester

BACHELOR OF TECHNOLOGY
IN
Artificial Intelligence & Machine Learning

Submitted By,
SREERANJINI P
01SU24AI102

UNDER THE GUIDANCE OF
Prof. Mahesh Kumar V B

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

SRINIVAS UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY
MUKKA, MANGALURU-574146

2025-26

SRINIVAS UNIVERSITY
INSTITUTE OF ENGINEERING & TECHNOLOGY

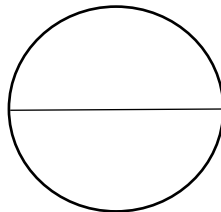


Department of Artificial Intelligence & Machine Learning

CERTIFICATE

This is to certify that, **Sreeranjini P** (**01SU24AI102**), has satisfactorily completed the assessment (Individual Task Report) in **Artificial Neural Networks (24SBT113)** prescribed by the Srinivas University for the 4th semester B. Tech course during the year **2025-26**.

MARKS AWARDED



Staff In charge

Prof. Mahesh Kumar V B



TABLE OF CONTENTS

Sl.no	Description	Page no.
1	Introduction	1
2	Passive Decay Model	3
3	Modified Passive Decay Model	3
4	Additive Autoassociative Model	4
5	Heteroassociative Model	5
6	Shunting Activation Model	6
7	Conclusion	7
8	References	8

Introduction to Neural Network Models

Artificial neural networks are computational models inspired by the structure and functioning of the human brain. They consist of interconnected units called neurons that process information through weighted connections, mimicking the way biological neurons communicate. Over the decades, various neural network models have been developed to solve specific computational problems, ranging from memory storage and pattern recognition to associative mapping and dynamic signal processing. Among these, five important models stand out: the Passive Decay Model, Modified Passive Decay Model, Additive Autoassociative Model, Heteroassociative Model, and Shunting Activation Model. The Passive Decay Model is one of the simplest continuous-time neural network models. In this model, the activation of each neuron naturally decreases over time unless it is stimulated by other neurons or external input. This decay behavior captures the fading nature of memory and neural signals in biological systems. While effective for modeling temporal dynamics, the simple linear decay limits its capacity to represent complex, nonlinear interactions. To address this limitation, the Modified Passive Decay Model was introduced. This model extends the basic passive decay concept by incorporating nonlinear activation functions, thresholds, or saturation limits. These modifications ensure that neuron activations remain within a plausible range and improve the stability and robustness of the network. The modified model is particularly useful for simulating more realistic biological neural behavior and for modeling tasks where nonlinear interactions between neurons are important. The Additive Autoassociative Model is designed for storing and recalling patterns within a single neural layer. In this network, neurons are interconnected with symmetric weights, allowing the system to retrieve complete patterns even when presented with partial or noisy inputs. This capability makes the autoassociative network ideal for memory based tasks, such as the Hopfield network, where the goal is to stabilize the network at a learned pattern. Its main strength lies in pattern completion, but it is

limited to linearly separable or stable patterns. In contrast, the Heteroassociative Model is designed for mapping input patterns to entirely different output patterns. Unlike autoassociative networks, the input and output layers are distinct, allowing the network to learn complex associations, translations, or mappings between sets of data. This makes heteroassociative networks suitable for supervised learning tasks, such as translating sensory input into motor output or associating stimuli with responses. Finally, the Shunting Activation Model introduces bounded and controlled dynamics to neuron activations. By separating excitatory and inhibitory influences and imposing upper and lower activation limits, this model prevents uncontrolled growth of neural activity and ensures stable behavior over time. The shunting mechanism is particularly effective for simulating realistic biological neural networks where activations must remain within physiological limits. It combines features of both continuous-time dynamics and nonlinear regulation, making it a versatile tool in computational neuroscience. Together, these five models provide a spectrum of neural network approaches, each addressing different computational challenges. From simple decay-based dynamics to pattern storage, associative mapping, and controlled activations, they demonstrate how network structure, weight design, and activation rules influence learning, memory, and signal processing. Understanding these models is essential for both theoretical and applied research in neural computation, cognitive modeling, and artificial intelligence.

1. Passive Decay Model

Purpose The Passive Decay Model is one of the simplest continuous-time neural network models. Its primary purpose is to simulate how a neuron's activation naturally decreases over time in the absence of external or internal stimulation. This mimics the biological behavior of neurons, where a signal gradually fades if not reinforced. It is widely used in modeling memory fading, short-term neural responses, and dynamic systems that require decaying signals. **Mathematical**

Representation The model is described by the differential equation: $\frac{dx_i}{dt} = -\lambda x_i + \sum_j w_{ij} y_j + I_i$ Where: x_i = activation of neuron i at a given time t λ = passive decay rate; a higher value means the neuron's activity fades faster w_{ij} = weight of the connection from neuron j to neuron i y_j = output of neuron j (can be the same as its activation or passed through an activation function) I_i = external input to neuron i

Breaking it down:

1. Decay term ($-\lambda x_i$):
 - o Represents the natural tendency of the neuron to lose activation over time.
 - o If there is no input ($y_j = 0$ and $I_i = 0$), the activation x_i will eventually decay to zero. This models "forgetting" or fading memory.
2. Summation term ($\sum w_{ij} y_j$):
 - o Represents the influence of other neurons in the network.
 - o If other neurons are active and have strong weights (w_{ij}), they can counteract decay and increase x_i .
3. External input (I_i):
 - o Represents stimuli coming from outside the network.
 - o Provides a way to "boost" neuron activity even if internal connections are weak.

Characteristics

- Continuous Decay: Without external or internal stimulation, each neuron's activation decreases exponentially toward zero.
- Dependent on Network Inputs: The actual behavior of a neuron depends not only on its decay rate but also on the input it receives from connected neurons and the environment.
- Linear Dynamics: The original passive decay model is linear, meaning the relationship between inputs, weights, and activation is straightforward.
- Simple Yet

Foundational: Despite its simplicity, this model is the foundation for more advanced dynamic models, such as modified passive decay or shunting models. Example Scenario Imagine a neuron that receives no input ($I_i = 0$) and has a decay rate $A = 0.5$. If its initial activation is $x_i = 1$: $\frac{dx_i}{dt} = -0.5 \cdot 1 = -0.5$ • After 1 second, the neuron's activation reduces to 0.5. • Without any input, it continues to decay toward 0. Now, if it receives a constant input $I_i = 0.5$, the decay slows down or even stabilizes at a non-zero activation level. This demonstrates how decay and input compete to determine the neuron's activity.

2.Modified Passive Decay Model

Purpose The Modified Passive Decay Model improves on the basic passive decay model by incorporating nonlinear activation functions, thresholds, or saturation limits. This makes the network more biologically realistic and stable, preventing activations from growing without bound or decaying unrealistically. It is often used in modeling memory retention, cognitive processes, and neural dynamics where nonlinear responses are observed. Equation $\frac{dx_i}{dt} = -Ax_i + \sum_j w_{ij} f(x_j) + I_i$ Where: • $-Ax_i$ • $\sum_j w_{ij} f(x_j)$ • I_i • $f(x_j)$ = nonlinear activation function (e.g., sigmoid, tanh, or step) • Other terms same as the passive decay model Characteristics • • • Nonlinear Response: The output is bounded (e.g., between 0 and 1), preventing unrealistic activations. Stability: The network avoids uncontrolled growth of activation values. Memory Modeling: Can represent neurons that respond strongly to input but gradually saturate. Example Suppose a neuron receives strong input from connected neurons. Without modification, activation might grow indefinitely. With a sigmoid activation function, the output saturates at 1, making the network stable.

3. Additive Autoassociative Model

Purpose This model is designed to store and recall patterns within a single layer of neurons. It's called "autoassociative" because the network can recall a complete pattern when presented with only part of it or a noisy version. Common in Hopfield networks, it's used for pattern recognition and memory tasks. Equation Where: •

• $i(t+1) = f(\sum_j w_{ij} x_j(t))$ • $i(t)$ = activation of neuron i at time t • w_{ij} = weight connecting neuron i and j (symmetric: $w_{ij} = w_{ji}$) • f = activation function (step or sigmoid)

Characteristics • **Pattern Completion:** Can reconstruct full patterns from partial inputs. • • • **Symmetric Weights:** Ensures network convergence to stable patterns. **Single Layer:** Input and output neurons are the same. **Discrete Updates:** Activations update in steps, typically asynchronously or synchronously. **Example** Given a binary pattern $[1, 0, 1]$ stored in the network, if only $[1, 0, ?]$ is presented, the network can fill in the missing value correctly, demonstrating associative memory.

4. Heteroassociative Model

Network Purpose Heteroassociative networks map input patterns to different output patterns. Unlike autoassociative models, the input layer is distinct from the output layer, making it suitable for pattern translation, supervised learning, or stimulus-response mapping. Equation $k = f(\sum_i w_{ki} i)$ Where:

• • • i = input neuron i • k = output neuron k • w_{ki} = weight connecting input i to output k • f = activation function

Characteristics • • • **Input-Output Mapping:** Can associate one set of patterns with another. **Supervised Learning:** Trained with target outputs. **Many-to-Many Associations:** Can handle complex mappings between layers. **Example** A network that converts letters into Morse code could be implemented as a heteroassociative model: input $[A] \rightarrow$ output $[\cdot -]$.

5. Shunting Activation Model

Purpose The Shunting Activation Model introduces bounded dynamics, ensuring that neuron activations stay within physiological limits. It separates excitatory and inhibitory influences, making it more realistic for biological neural networks and continuous time computations. Equation $\frac{dx_i}{dt} = -Ax_i + (B - x_i) \sum_j w_{ij}^+ y_j + x_i \sum_j w_{ij}^- y_j - I_i$

Where: x_i = activation of neuron i • • • B = upper bound of activation A = lower bound of activation w_{ij}^+ = excitatory weights w_{ij}^- = inhibitory weights I_i = external input •

Characteristics • • • **Bounded Activation:** Prevents runaway neuron activity. **Excitation vs Inhibition:** Models the balance between stimulatory and inhibitory inputs. **Continuous Dynamics:** Useful for simulations where time evolution of activation matters. **Example** If a neuron receives strong excitatory input but is near its upper bound B , further input has limited effect due to shunting, keeping the network stable.

Conclusion

Neural network models provide a versatile framework for understanding how information can be processed, stored, and recalled, both in artificial systems and biological networks. The models studied—Passive Decay, Modified Passive Decay, Additive Autoassociative, Heteroassociative, and Shunting Activation—each offer unique mechanisms for handling neuron activation, learning, and memory. The Passive Decay Model demonstrates how activations naturally fade over time, forming the foundation for dynamic neural simulations. The Modified Passive Decay Model builds upon this by introducing nonlinear activation functions, improving stability and biological realism. The Additive Autoassociative Model emphasizes pattern storage and retrieval, showcasing the network's ability to complete partial or noisy patterns. In contrast, the Heteroassociative Model extends this concept to map inputs to distinct outputs, making it suitable for supervised learning and associative tasks. Finally, the Shunting Activation Model introduces bounded activations and separate excitatory/inhibitory influences, offering a more physiologically accurate representation of neuron behavior. Collectively, these models highlight the importance of network structure, weight configuration, and activation rules in determining learning capability and stability. Understanding these models is essential not only for artificial intelligence applications, such as pattern recognition and classification, but also for modeling cognitive processes and

neural dynamics in the brain. Studying these models provides a strong foundation for exploring more advanced neural networks and deep learning architectures.

References

1. Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8), 2554–2558.
2. Rosenblatt, F. (1958). The perceptron: A probabilistic model for information storage and organization in the brain. *Psychological Review*, 65(6), 386–408.
3. Haykin, S. (2009). *Neural networks and learning machines* (3rd ed.). Pearson Education.
4. Anderson, J. A. (1995). *An introduction to neural networks* (2nd ed.). MIT Press.
5. Brownlee, J. (2016). *A gentle introduction to the perceptron algorithm for machine learning*. Machine Learning Mastery. Retrieved from <https://machinelearningmastery.com/a-gentle-introduction-to-the-perceptron-algorithm-for-machine-learning/>
6. Hertz, J., Krogh, A., & Palmer, R. G. (1991). *Introduction to the theory of neural computation*. Addison-Wesley.